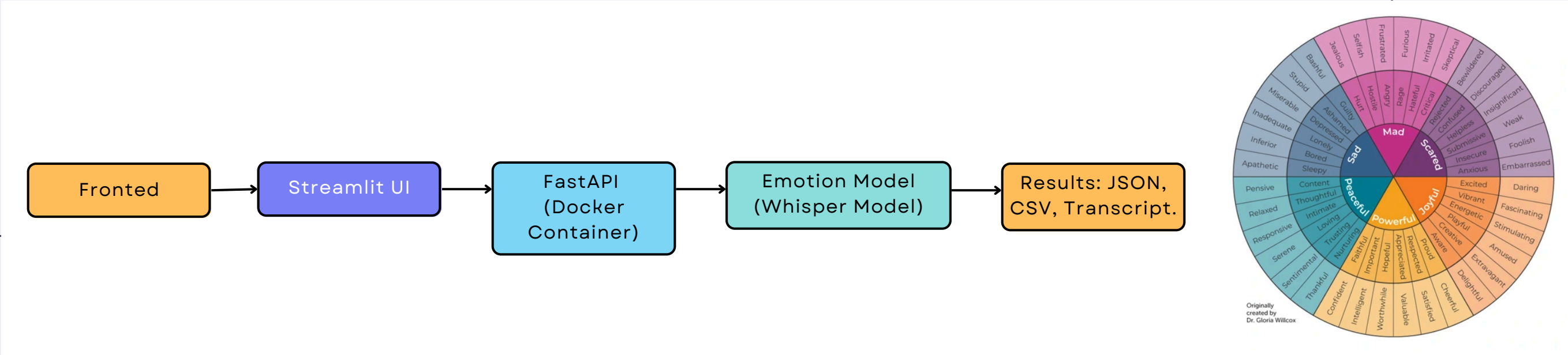


Emotion Recognition API with Dockerized Deployment & Frontend

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A scalable API for emotion detection and audio transcription, using FastAPI, Transformers, Whisper, and Docker. Batch and real-time predictions, with an additional Streamlit frontend for interactive demos. Production practices applied: CI, unit testing, logging, stress testing.



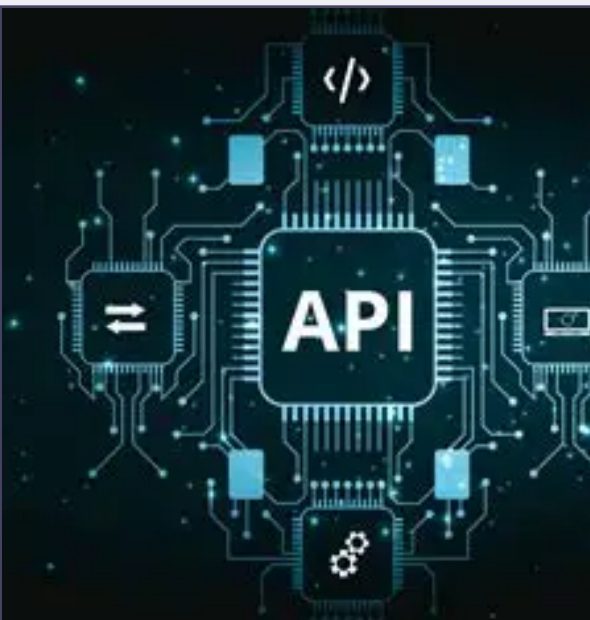
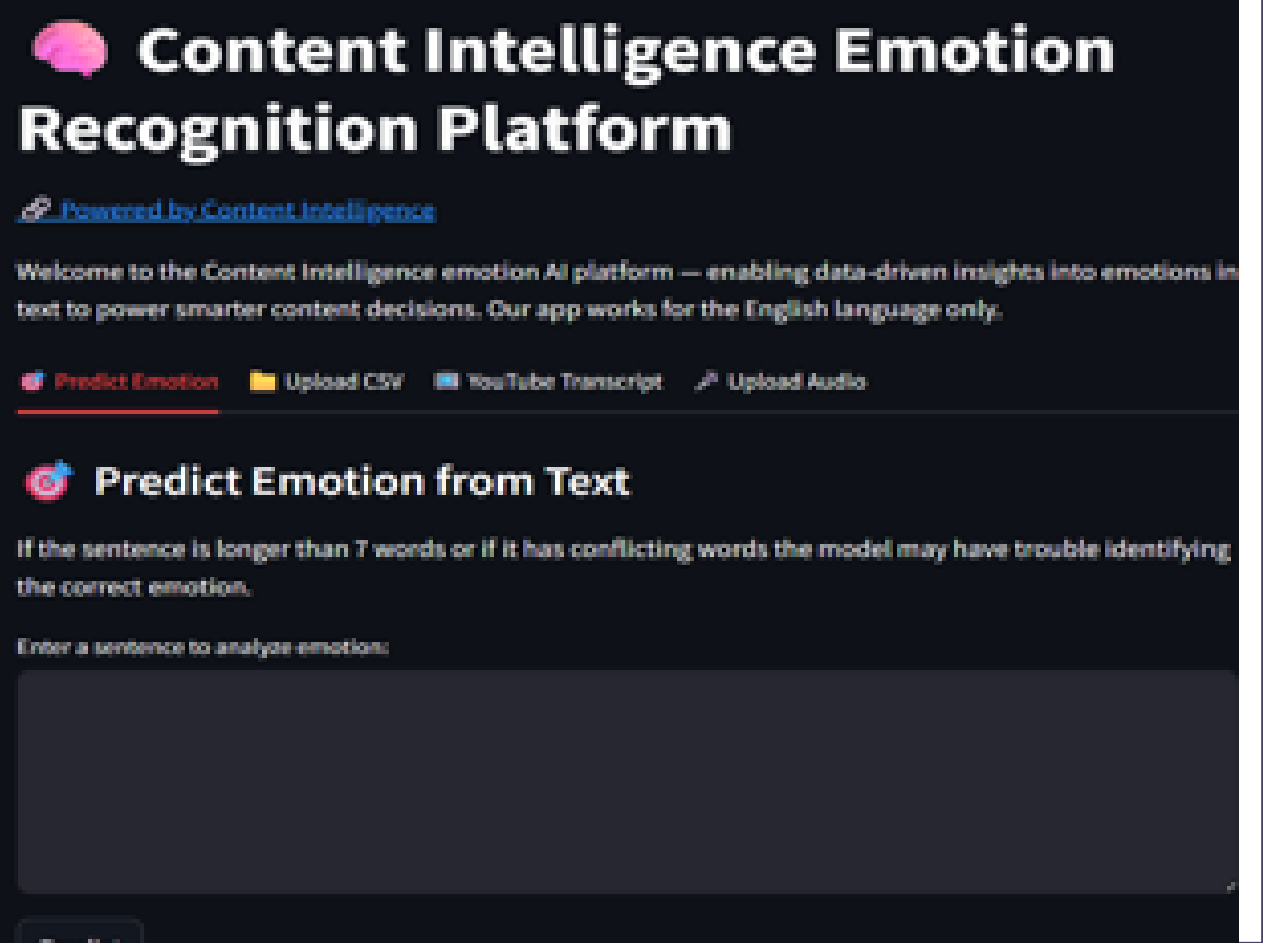
- Text & Batch Emotion Prediction: via `/predict` & `/upload-csv`
- YouTube & Audio Transcription: audio-to-text, then batch prediction
- Dockerized for Easy Deployment: runs anywhere, “works on my machine”
- Logging & Unit Tests: robust and maintainable code
- Streamlit Frontend: for simple web-based demo (built, attempted Azure deploy)

- Python
- FastAPI
- Hugging Face Transformers
- Whisper
- Docker
- Streamlit
- Locust
- Pandas

Locust Results:

- 1,978 requests (POST /predict) at up to 44.7 RPS
- Avg. response: 117 ms | Median: 90 ms | 99%ile: 270 ms
- Failures: 1 (0.1% error)
- Conclusion: Stable under high concurrent load; near real-time for typical use.

- User-friendly interface built in Streamlit for uploading text, audio, or CSV files and viewing predictions.
- “Designed for ease of use and future deployment. Deployment to Azure not fully functional at demo time, but local interface works.”



Endpoint	Description
/predict	Predict emotion from text
/upload-csv	Batch predict from CSV
/youtube-transcript	Get transcript from Youtube Audio
/upload-audio-transcript	Transcript from audio file

